

XV6 Problem Set #1 & Assignment #1

Execution Screenshots

Anser Abbas
ERP: 31675

xv6-labs-2026 - Unix Utilities Lab

Problem Set #1 - sleep and sixfive

Execution of sleep and sixfive in xv6, with identification visible.

```
xv6-labs-2026 — qemu-system-riscv64 ◀ make qemu — 80x24

s user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan
n user/_dorphan user/_sync user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
balloc: first 1185 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
[$ echo Anser Abbas - ERP 31675 ]
Anser Abbas - ERP 31675
[$ sleep 10 ]
[$ echo 5 7 12 25 13 18 | sixfive ]
5
12
25
18
$ █
```

Problem Set #1 - memdump

Execution of memdump showing all example outputs.

```
xv6-labs-2026 — qemu-system-riscv64 ◀ make qemu — 80x24

[$ echo Anser Abbas — ERP 31675 ]
Anser Abbas — ERP 31675
[$ memdump ]
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
CE8
1819438967
100
z
xyzy
Example 5:
hello
w
o
r
l
d
$ █
```

Problem Set #1 - find and find -exec

Execution of recursive find and find with -exec.

```
xv6-labs-2026 — qemu-system-riscv64 ◀ make qemu — 80×24

user/_forktest user/_sleep user/_sixfive user/_memdump user/_find user/_grep use
r/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressf
s user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forpha
n user/_dorphan user/_sync user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
balloc: first 1185 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
[$ echo Anser Abbas - ERP 31675 ]
Anser Abbas - ERP 31675
[$ find . README ]
./README
[$ find . wc -exec echo found ]
found ./wc
$ █
```

Assignment #1 - uptime and regex find

Execution of uptime and regular-expression matching in find.

```
xv6-labs-2026 — qemu-system-riscv64 ◀ make qemu — 80x24

ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
[$ echo Anser Abbas - ERP 31675 ]
Anser Abbas - ERP 31675
[$ uptime ]
98
[$ echo > test1 ]
[$ echo > test2 ]
[$ find . test. ]
./findtest.sh
./usertests
./test1
./test2
[$ find . ^test1$ ]
./test1
$ █
```

Assignment #1 - shell improvement

Shell reading commands from redirected file input without printing an extra prompt.

```
xv6-labs-2026 — qemu-system-riscv64 ◀ make qemu — 80x24

mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo
user/_forktest user/_sleep user/_pingpong user/_sixfive user/_memdump user/_find
user/_uptime user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir use
r/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sync user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
ballocc: first 1260 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
[$ echo Anser Abbas - ERP 31675 ]
Anser Abbas - ERP 31675
[$ echo echo hello > script ]
[$ sh < script ]
hello
$ █
```